

AI/ML INTERNSHIP PROGRAM

5-Day Project Brief

02

Law Firm Document Review Bottleneck

AI-Powered Legal Due Diligence System · RAG Architecture

DOMAIN AI / ML	TEAM Pair A	DURATION 5 Days	USE ALL FREE RESOURCES
--------------------------------------	-----------------------------------	---------------------------------------	-----------------------------------

TARGET INDUSTRY

Corporate & Real Estate Law Firms · Lahore, Karachi, Islamabad

SECTION 1: PROBLEM STATEMENT

The Challenge

Mid-size law firms across Pakistan — particularly those operating in corporate law, real estate transactions, and financial advisory — face a significant operational bottleneck in their document review and due diligence processes. When a high-value transaction arrives, such as a property acquisition, merger, or loan agreement, junior associates are tasked with manually reviewing large bundles of legal documents, which may run into hundreds of pages, in order to identify specific clauses, flag inconsistencies, locate missing information, and produce a structured review memorandum for senior partners.

This process is inherently slow, resource-intensive, and prone to human error. Junior lawyers are often inconsistent in what they flag, frequently miss subtle contradictions across documents, and produce review memos of uneven quality. Senior partners must then spend considerable time correcting and augmenting this work — time that carries a high opportunity cost given their billable rates.

CORE PAIN POINT

A typical due diligence review involving 30–50 pages of legal documents currently takes a junior associate team 5 to 7 working days to complete. Clients, however, demand turnaround within 48 hours for competitive transactions. This gap in delivery speed is directly causing client dissatisfaction and, in some cases, loss of mandates to larger firms with greater associate bandwidth.

Why This Problem Is Specific to Pakistan

The legal technology landscape in Pakistan remains nascent. Unlike law firms in the United Kingdom or the United States where AI-assisted due diligence tools from vendors such as Kira Systems and Luminance are now standard, Pakistani law firms have virtually no access to locally configured, affordable, and contextually appropriate legal AI tooling. Document formats, local statute references, and Urdu-language clauses embedded within otherwise English documents make off-the-shelf international solutions poorly suited to the Pakistani legal environment.

Furthermore, the economics of mid-size Pakistani law firms — typically five to thirty lawyers — do not support the PKR 500,000+ monthly licensing costs of enterprise legal AI platforms. This creates a clear market gap that a purpose-built, locally configured solution can fill at a fraction of the cost.

Who Is Affected

5–7 Days Current Due Diligence Turnaround	PKR 250K/wk Senior Associate Time on Low-Value Review	48 Hours What Clients Actually Expect
---	---	---

Primary Stakeholders

- Managing Partner of a 5–30 lawyer corporate or real estate law firm in Lahore, Karachi, or Islamabad
- Legal Head at a bank or financial institution responsible for reviewing loan and security documentation

- Senior Associates who spend the majority of their time on first-pass document review rather than higher-value legal reasoning
- Clients (businesses, developers, financial institutions) who lose competitive advantage due to slow due diligence timelines

Cost of Inaction

Findout

SECTION 2: PROPOSED SOLUTION — MVP DEFINITION

Solution Overview

The solution is a Retrieval-Augmented Generation (RAG) document review system specifically configured for Pakistani legal due diligence workflows. The system allows a law firm to upload a bundle of legal documents — in PDF format — through a simple web interface. The system then indexes these documents using vector embeddings, applies a pre-defined review checklist appropriate to the transaction type (property due diligence, loan agreement review, or company acquisition), and automatically generates a structured review memorandum highlighting clause-by-clause findings, identified red flags, and flagged missing documents.

The lawyer's role shifts from spending two days writing the review from scratch to spending two hours reviewing, correcting, and finalising an AI-generated first draft. This compresses the due diligence turnaround from five to seven days down to same-day delivery for standard transactions.

Technical Architecture

Technology Stack

Pricing Model

SECTION 3: COST ANALYSIS

SECTION 4: 5-DAY SUBMISSION SCHEDULE

Each day below specifies exactly what must be completed, demonstrated, and submitted by end of day. Work is performed in Pair A (two interns collaborating on all tasks).

**DAY
1****Problem Design & Solution Architecture****Submission Requirements:**

- Written Problem Statement (this document, annotated with pair's own research notes)
- Architecture Diagram: data flow from PDF upload → embedding → query → memo output
- Technology selection rationale: why LlamaIndex / LangChain / Claude API
- Sample due diligence checklist (10–15 questions for property transaction review)
- Cost breakdown spreadsheet with API pricing estimates
- Identification of at least 2 target law firms in Rawalpindi/Islamabad for Day 4 demo

Key Deliverable:

Signed-off architecture diagram + cost sheet submitted to supervisor before end of Day 1.

**DAY
2****Core Pipeline Build — Document Ingestion & Embedding****Submission Requirements:**

- Working PDF ingestion module (accepts multi-document bundle, handles scanned PDFs via OCR fallback)
- Vector embedding pipeline operational (chunking strategy documented — recommend 512 token chunks with 50 token overlap)
- LlamaIndex index created and queryable from command line
- Test run: upload a 20-page sample property agreement and confirm index created successfully
- Code pushed to shared repository with README

Key Deliverable:

Supervisor test: upload a provided sample document and confirm the system indexes it without errors.

**DAY
3****Query Engine & Memo Generation — Complete Build****Submission Requirements:**

- Checklist query engine integrated: all 10–15 checklist questions run automatically against the indexed document bundle
- Claude API integration complete: structured memo generated in Word / PDF format
- React frontend operational: upload interface → processing status → memo download
- Red flag detection working: at least 3 defined red flag patterns flagged correctly
- End-to-end test: full document bundle in → structured review memo out, in under 5 minutes
- Internal demo to supervisor (30-minute walkthrough)

Key Deliverable:

Full end-to-end system demo to supervisor. System must produce a readable, structured memo from a real document bundle.

**DAY
4****Client Customisation & Potential Client Run****Submission Requirements:**

- System customised for first target client: their letterhead on output memo, their specific checklist (property / loan / acquisition — confirm with client)
- Live demo conducted with at least one target law firm (in-person or video call)
- Demo script prepared and rehearsed: document bundle uploaded live, memo generated in front of client
- Client feedback documented: what they liked, what they requested to change
- Proposal document prepared: pricing, onboarding timeline, what is included

Key Deliverable:

Completed demo with a real law firm contact. Client feedback form filled and submitted.

**DAY
5****Final Run, Refinement & Second Client Outreach****Submission Requirements:**

- Refinements from Day 4 feedback implemented (within scope — no new features, only corrections)
- Second client demo conducted or scheduled (from the provided potential client list)
- Final system documentation: setup guide, user manual (1–2 pages), known limitations
- Final presentation prepared (10 slides max): problem, solution, demo screenshots, pricing, next steps
- Presentation delivered to supervisor and team
- Repository finalised, code commented, environment variables documented

Key Deliverable:

Final presentation delivered. Repository submitted. At least one client demo completed, second demo scheduled.